APPENDIX V. MALAWI—JOINT IMF/WORLD BANK DEBT SUSTAINABILITY ANALYSIS[8]

Based on the Joint Fund-Bank Debt Sustainability Framework for Low-income Countries (LIC DSA), Malawi's risk of external debt distress remains moderate. Malawi's narrow export base, reliance on rain-fed agriculture, and weak international reserves leave it vulnerable to terms-of-trade, and weather shocks. Stress testing reveals potential vulnerabilities in Malawi's public debt situation. In particular, the external sector debt burden indicator (PV of debt-to-exports) breaches the sustainability threshold after an export shock. Malawi's debt situation is expected to improve over the medium- to long- run, reflecting strong economic growth under a floating exchange rate regime and a liberalized current account. However, measures to support a diversification of the export base will also be important. Fiscal dominance has contributed to the increase in domestic debt burden indicators, whereby the PV of public debt to GDP ratio reaches about 45.5 percent in 2012 and gradually comes down to about 13.9 percent by 2032. Stress tests to public sector debt dynamics reveal the need for significant fiscal consolidation and reform of parastatal institutions.

I. BACKGROUND

20. **Malawi's medium- and long-term public and publicly guaranteed (PPG) external debt is estimated at US$1.14 billion, reflecting a significant increase in recent years.** Malawi's external debt was lowered to a sustainable level of about 15 percent of GDP in 2006, after it received debt relief under the Heavily Indebted Poor Countries (HIPC) initiative and the Multilateral Debt Relief Initiative (MDRI). The level of the external debt stock has increased substantially since then, to 23 percent of GDP (about US$1.14 billion) at end-2011. The share of multilateral debt decreased (to about 73 percent from 90 percent as at end-2008). Malawi's main bilateral creditors are China and India, while the main multilateral creditor is IDA. In addition, increased government expenditure at a time when external grants fell sharply and revenue performance deteriorated, contributed to the build-up of the domestic debt in 2011 (to about 25 percent of GDP from 20 percent of GDP as at end-2008). The level of total public debt in Malawi is estimated at US$2.4 billion. Reliable figures on private external debt are not available, although the amounts are not believed to be large.

[8] The DSA was prepared by IMF and World Bank staff in collaboration with the Malawian authorities. The analysis updates the previous Joint DSA dated February 10, 2010 (IMF Country Report No. 10/24). The DSA follows the IMF and World Bank Staff Guidance Note on the Application of the Joint Fund-Bank Debt Sustainability Framework for Low-Income Countries (January 22, 2010).

Malawi: Stock Medium- and Long-Term External Public and Publicly Guaranteed Debt (DOD) as of end-2011

	US$ Million	Share
Grand Total	1140.49	100.0
Multilateral		
IDA	275.97	24.2
ADF	161.10	14.1
IMF	146.00	12.8
EIB	21.41	1.9
IFAD	79.42	7.0
NDF	30.02	2.6
BADEA	31.90	2.8
OPEC	27.27	2.4
PTA	70.00	6.1
Total Multilateral	843.09	73.9
Bilateral		
Kuwait	43.90	3.8
Taiwan	13.00	1.1
India	71.70	6.3
Belgium	2.20	0.2
China	154.80	13.6
France	11.80	1.0
Total Bilateral	297.40	26.1

Sources: Malawian authorities and IMF staff estimates.

II. UNDERLYING DSA ASSUMPTIONS

21. **The baseline scenario reflects the recent policy measures taken by the authorities to restore macroeconomic stability in Malawi.** In May 2012, the authorities implemented a set of measures to address Malawi's chronic balance of payments difficulties and to halt a slowdown in economic activity due to shortages of foreign exchange and critical imports. The policies included a devaluation of the exchange rate from K167 to K250 per U.S. dollar in May 2012, and adoption of a floating exchange rate regime. Further policy measures are envisaged under a new Extended Credit Facility (ECF) arrangement, including prudent fiscal and monetary policies, structural reforms and policies to enhance the investment climate, promote export diversification and foster growth. The baseline assumes a gradual reduction in the external current account deficit through export diversification, and reliance on grants and concessional financing in the medium term. The current mining projects are expected to continue as planned combined with additional projects in uranium and other minerals such as niobium. It also assumes that contingent risks stemming from parastatal organizations are contained. The key macroeconomic assumptions are summarized in Box 1.

22. **The evolution of key debt indicators followed a somewhat different trajectory than the projections in the previous DSA (February 2010).** For example, the ratio of total public debt to GDP was 48 percent at end-2011, compared with a projected level of

34 percent, reflecting higher-than-projected levels of domestic borrowing. By contrast, the PV of external debt-to-export ratio, estimated at 61 percent for end-2011, was slightly lower than projected (65 percent), on account of better-than-expected export performance. The annual average real GDP growth for 2009–11 was in line with the projected 6.6 percent, but there was a marked downward trend compared to a steady path in the projections. Going forward, the current DSA projects long-term growth of 6 percent per year, compared to 6.7 percent in the previous DSA. The current projection is consistent with the historical average growth rate of 5.8 percent over the last ten years.

23. **Malawi's economic performance is expected to improve over the medium term.** Real GDP growth is projected to rise gradually from 4.3 percent in 2012 to about 6½ percent over the medium term. The floating of the kwacha and liberalization of the foreign exchange regime is expected to foster growth over the medium term by redirecting resources to exports and import-competing production, while containing import demand for consumption goods. In the near term, growth will be generated by the sectors that were hardest hit by the foreign exchange shortages, including manufacturing, transportation service, construction, and trade, as production levels increase toward capacity. Other sectors, including agriculture, will also benefit from the easing of fuel shortages. Over time, growth is expected to be led by mining and non-traditional agricultural exports. Following the recent devaluation of Kwacha, a spike in inflation is expected in the next few months (reaching about 20 percent by end-2012), but this should be reversed with implementation of restrained fiscal and monetary policies over the medium term.

24. **The main downside risks to the debt sustainability outlook are deterioration in the terms of trade and adverse weather conditions that lower agricultural output.** Malawi's narrow export base, reliance on rain-fed agriculture, and weak international reserves leave it vulnerable to terms-of-trade and weather shocks. Stress tests (see next section) show that the main external sector debt burden indicator (PV of debt-to-exports ratio) breaches the sustainability threshold after an export shock—a proxy for both the terms-of-trade and weather shocks. Malawi's debt situation is expected to improve over the medium-to-long run. The recent liberalization of the foreign exchange regime for current account transactions is expected to encourage private investment and promote diversified growth. Repairing strained relations with donors and more consistent implementation of prudent fiscal and monetary policies are expected to increase foreign exchange inflows into the country and facilitate the accumulation of international reserves.

[9] Malawi's stock of external debt declined substantially from about 112 percent of GDP in 2005 to around 14.6 percent in 2006. The previous DSA estimated the stock of Malawi's debt at US$683 million as of end-2008.

Box 1: Baseline Macroeconomic Assumptions

Real GDP is projected to grow by about 6 percent per year on average over the long term, owing to timely adoption of a comprehensive policy package by the authorities. The main drivers of growth will be agriculture, manufacturing, trade, and mining sectors. The adopted policy package is underpinned by foreign exchange market reforms and structural measures to enhance the productive capacity of the economy.

The mining sector is projected to increase its contribution to GDP. New mining operations will include the production of rare earth metals (e.g., ferroniobium, tantalum fluorite, and zircon).

Consumer price inflation is projected to increase to an annual average rate of 18.4 percent in 2012 after the devaluation in May 2012. However, inflation is projected to decelerate in the medium term after a short-lived spike, reaching single-digit levels. The nominal effective exchange rate is assumed to be determined by inflation differential between Malawi and its trading partners after 2014.

Revenue (excluding grants) is projected to increase relative to GDP in FY2012/13 owing to the positive impact of a unified exchange rate regime on tax revenues from international trade by increasing the share of formal transactions being taxed and the adoption of an automatic fuel price adjustment mechanism. In subsequent years, both tax and non-tax revenues improve gradually owing to improved tax efforts and expanded tax base. More fiscal discipline, entailing expenditure prioritization to protect critical social and infrastructure programs will be required. Domestic financing of the budget is projected to stay at zero percent of GDP in the medium term but will increase to an average 0.5 percent of GDP after FY2017/18.

Highly concessional financing from multilaterals is assumed to be the major source of debt (two-thirds), with somewhat less concessional financing from non-Paris Club creditors providing the balance. A small but increasing amount of borrowing on commercial terms is assumed for the outer years.

Grants are projected to increase to about 10.4 percent of GDP during FY 2012/13 as development partners are increasing their support for Malawi after the exchange rate unification. This reflects the restoration of budget support grants and increase in project grants. Afterwards, total grants are projected to stabilize in U.S. dollar terms but decrease gradually as a ratio of GDP. With increased support from donors, total external financing of government operations will also increase.

Current account deficit, including official transfers, is projected to narrow over the medium term. The exchange rate liberalization is expected to slow growth in imports (especially of consumer goods) and boost exports. Policy reforms are also expected to attract private capital inflows into Malawi and a gradual build up of international reserves.

Exports performance is projected to strengthen in the medium term as the change in exchange rate regime is expected to mobilize resources toward export-oriented activities. Export growth is projected to average about 9 percent during 2012–32. Tobacco's share in overall export is expected to decline as global demand weakens and the farmers diversify to other commodities. However, tobacco is assumed to be substituted by other non-traditional agricultural products over the medium term.

Imports grew at an average annual rate of 10 percent during 2002–11. The floating of the kwacha and liberalization of the foreign exchange regime is expected to contain imports demand over the medium term due to cost considerations and lower purchasing power of kwacha. Imports are projected to grow at an annual average rate of about 8 percent during the period 2012–32.

III. External Debt Sustainability

Baseline scenario

25. Under the baseline scenario and over the projection period, external debt burden indicators fall and remain below the indicative policy thresholds (Figure 1 and Table 1)[10].

[10] The World Bank's Country Policy and Institutions Assessment (CPIA) ranks Malawi as a medium performer (the average CPIA in 2008–10 is 3.37). Thus, the external debt burden thresholds for Malawi are (i) a PV of debt-to-GDP at: 40 percent; (ii) a PV of debt-to-exports at: 150 percent; (iii) a PV of debt-to-revenue at: 250 percent; (iv) a debt service-to-exports at: 20 percent: and (v) a debt service-to-revenue at: 30 percent.

The stock of external public and publicly guaranteed (PPG) debt is expected to gradually decline in the long term from about 30 percent of GDP in 2012 to about 14 percent in 2032. The present value (PV) of PPG external debt-to-GDP gradually declines to below 8.7 percent during the rest of the projection period. The PV of debt-to-exports ratio remains well below the threshold of 150 percent throughout the entire period. The PV of PPG external debt to revenue is also on a downward trend from a peak of 105 percent in 2012 and will remain well below the threshold of 250 percent during the entire period.

Stress tests

26. **The PV of debt-to-exports ratio (which is one of the external debt sustainability thresholds) is breached under stress tests (exports shock in particular).** The standard sensitivity analysis points to medium risk of debt distress (Figure 1 and Table 3), as one debt burden indicator (PV of debt-to-exports ratio) breaches its respective threshold after the shock in 2014 reaching the value of 185 (in percent of GDP) and only coming down to 138 percent of GDP in 2021. Based on the bound tests, the most extreme stress for the PV of debt-to-exports ratio is a one-standard deviation shock to the export value growth in 2013–14. Tobacco still accounts for more than half of Malawi's export earnings, and though this proportion is expected to decline over time, Malawi will remain vulnerable to a shock to tobacco export values for some time. Burley accounts for the majority of production, leaving Malawi vulnerable to changes in tastes and regulatory actions that could dramatically shrink what is already a stagnant global market.

27. **Debt service ratios are expected to remain below the threshold for the projection period.** All debt service ratios remain below their respective thresholds (Figure 1 and Table 1) both in the baseline and under the shock scenarios.

IV. Public Debt Sustainability

Baseline scenario

28. **Malawi's public debt has risen substantially since the run-up to the May 2009 presidential and parliamentary elections.** The share of domestic debt in the total public debt of Malawi is high (about 52 percent of total public debt) owing to increased domestic debt issuance by the government and the financing of the fiscal deficit by the central bank (Figure 2 and Table 2). The PV of public debt to GDP ratio is expected to reach about 45.5 percent in 2012 and only come down to the levels below 32 percent in the medium term and about 14 percent by 2032.

Stress tests

29. **Stress tests indicate that key ratios could deviate significantly from the baseline levels in the face of less favorable macroeconomic conditions (Figure 2 and Table 4).** Maintaining the primary balance at the 2012 level leads to a slight decline in

the PV debt-to-GDP ratio from 45 percent in 2012 to 41 percent at the end of the projection period (2032), compared to a decline to 14 percent under the baseline. A permanently lower GDP growth rate than the baseline produces a less pronounced decline in the PV debt-to-GDP ratio to 34 percent, compared to 14 percent under the baseline.

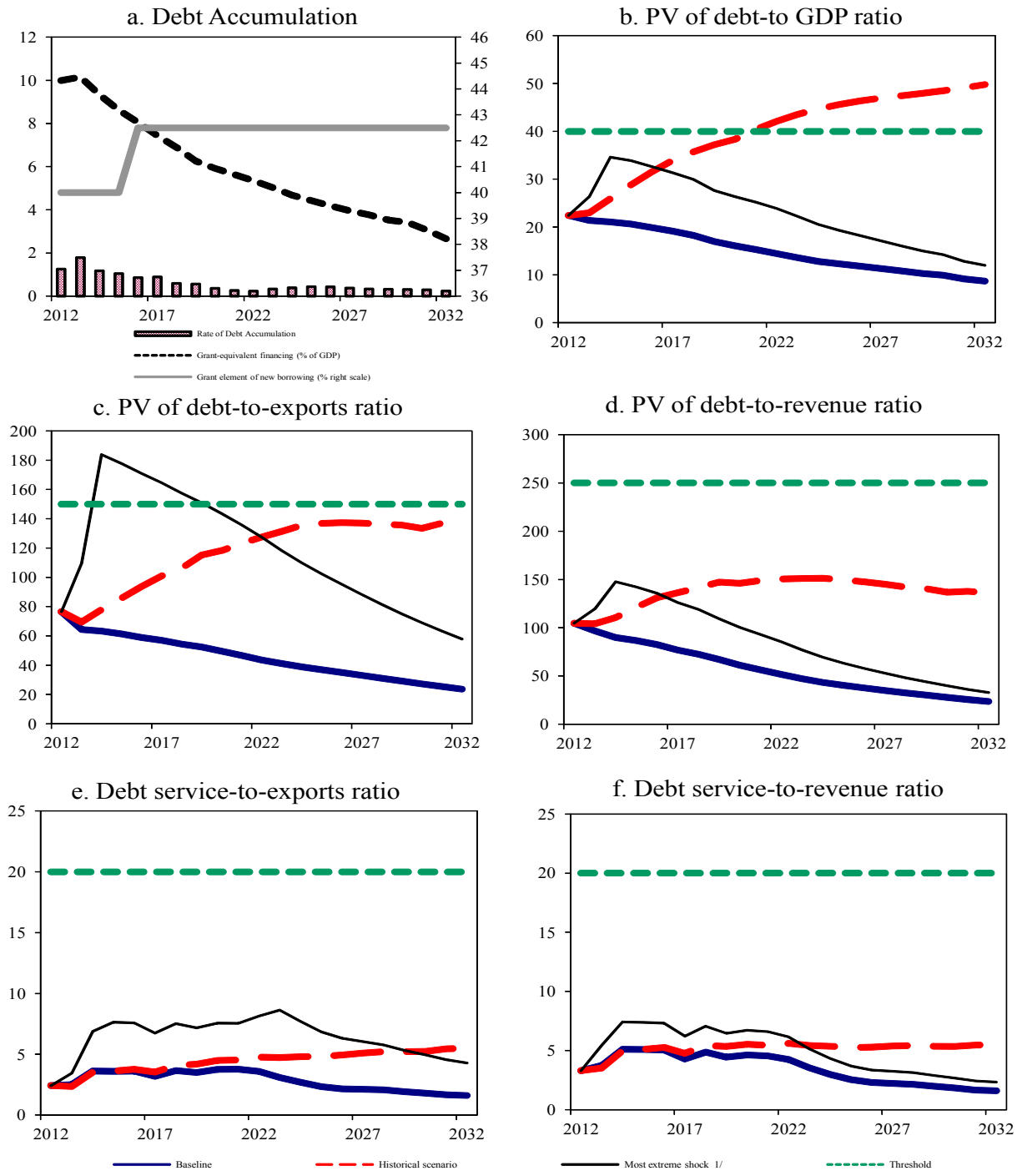
30. **Malawi is subject to a number of fiscal risks stemming from the activities of loss-making parastatal institutions.** The government's efforts to achieve fiscal sustainability and reduce the risks to the budget posed by contingent liabilities and operational losses of state owned enterprises are critical. In this connection, measures to enhance revenue performance, restrain growth in expenditures, limiting the National Oil Company to its core mandate of managing strategic reserves of fuel and a robust regulatory regime for public utilities that focuses on cost recovery, are very important.

IV. CONCLUSIONS

31. **Malawi is assessed to be at medium risk of debt distress.** The external debt indicator (PV of debt-to-exports) breaches its threshold in the scenario of bound tests. Stress tests to public sector debt dynamics underpin the need for effective fiscal consolidation, while protecting critical social and infrastructure expenditures.

32. **The adoption of corrective policy measures in 2012 has significantly improved the outlook for Malawi's debt sustainability, but risks related to terms of trade and weather shocks remain.** In particular, Malawi's heavy reliance on rain-fed agriculture and narrow export base make its export earnings vulnerable to fluctuations in world prices of its key exports. However, the liberalization of the foreign exchange regime will help to remove distortions and thereby encourage private investment and diversified growth, and help enhance the country's debt carrying capacity. Implementation of sound macroeconomic policies, diversification of the export base, and sustained growth would help reduce the risk of debt distress.

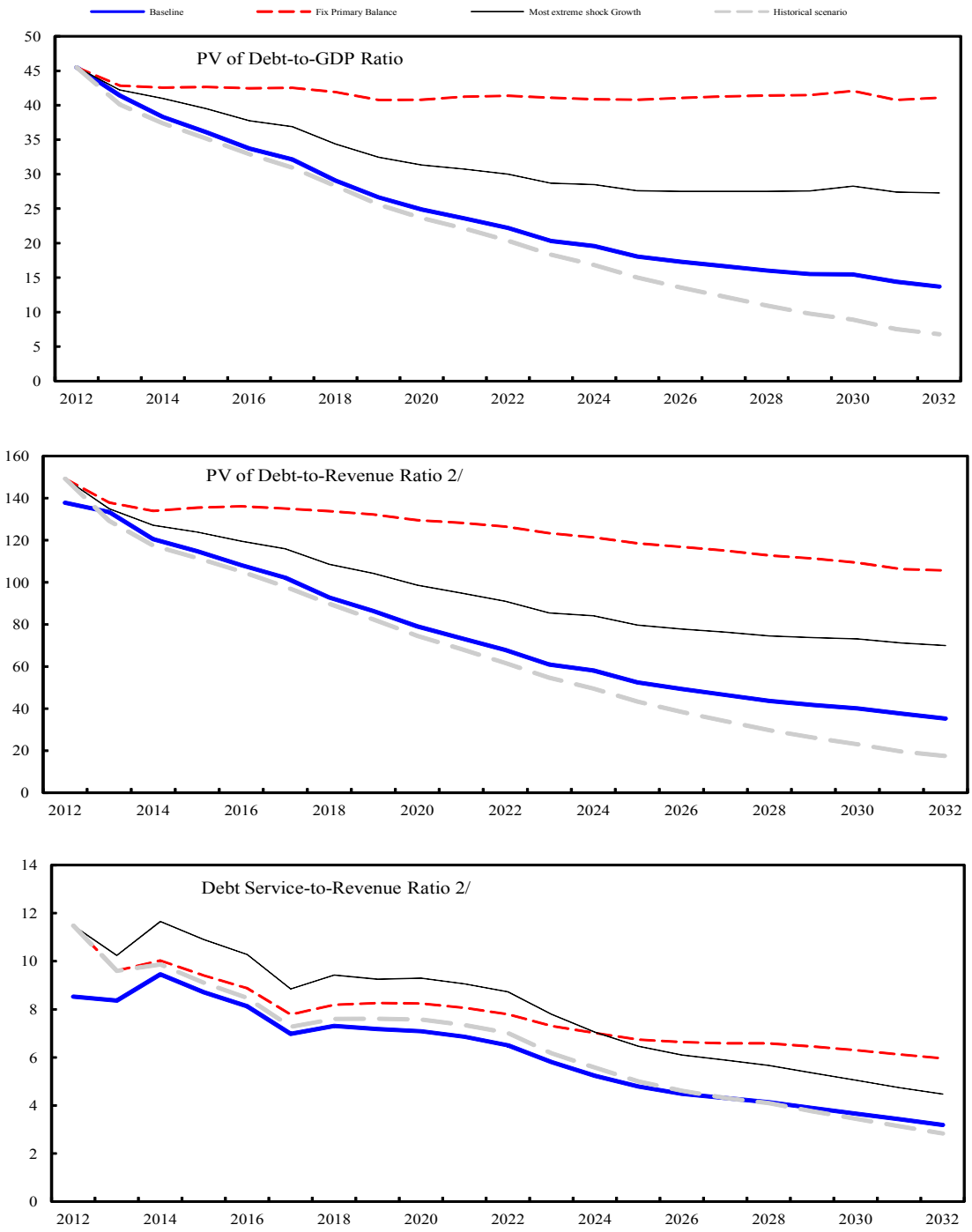
Figure 1. Malawi: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2012-2032 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2022. In figure b. it corresponds to a Exports shock; in c. to a Exports shock; in d. to a Exports shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2.Malawi: Indicators of Public Debt Under Alternative Scenarios, 2012-2032 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2022.
2/ Revenues are defined inclusive of grants.

Table 1.: External Debt Sustainability Framework, Baseline Scenario, 2009-2032 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections														
	2009	2010	2011			2012	2013	2014	2015	2016	2017	2012-2017 Average	2018	2019	2020	2021	2022	2031	2032	2018-2032 Average
External debt (nominal) 1/	**19.4**	**18.4**	**23.0**			**29.7**	**28.4**	**28.2**	**27.7**	**26.8**	**25.9**		**25.0**	**23.7**	**23.0**	**22.5**	**21.8**	**14.4**	**13.7**	
o/w public and publicly guaranteed (PPG)	19.4	18.4	23.0			29.7	28.4	28.2	27.7	26.8	25.9		25.0	23.7	23.0	22.5	21.8	14.4	13.7	
Change in external debt	-0.1	-1.0	4.6			6.7	-1.3	-0.2	-0.5	-0.9	-0.9		-0.8	-1.4	-0.6	-0.6	-0.7	-1.2	-0.7	
Identified net debt-creating flows	1.9	-1.1	4.3			1.3	-2.2	-2.5	-3.0	-3.1	-2.7		-2.1	-1.6	-1.2	-2.9	-2.6	-2.1	-2.5	
Non-interest current account deficit	**3.1**	**1.2**	**5.9**	**6.8**	**4.9**	**4.1**	**1.4**	**1.5**	**1.4**	**1.6**	**2.2**		**2.5**	**2.7**	**2.9**	**1.1**	**1.4**	**1.0**	**0.6**	1.8
Deficit in balance of goods and services	22.1	20.4	14.9			18.4	15.4	14.6	13.7	13.2	13.1		12.7	12.1	12.0	11.9	11.7	7.6	6.9	
Exports	24.6	27.0	26.1			29.3	33.2	33.3	33.6	33.7	33.6		33.6	32.3	32.4	32.8	33.1	35.8	36.6	
Imports	46.6	47.4	41.0			47.7	48.6	47.8	47.3	46.9	46.7		46.4	44.4	44.4	44.7	44.8	43.4	43.5	
Net current transfers (negative = inflow)	-17.0	-21.9	-10.9	-16.2	4.1	-17.1	-16.9	-16.1	-15.3	-14.6	-13.9		-13.1	-12.1	-11.7	-11.3	-10.9	-7.3	-7.0	-9.7
o/w official	-11.0	-16.9	-6.2			-11.5	-11.1	-10.3	-9.6	-8.9	-8.3		-7.5	-6.8	-6.5	-6.1	-5.8	-3.3	-3.1	
Other current account flows (negative = net inflow)	-2.0	2.7	1.9			2.8	2.9	3.0	3.0	2.9	3.0		2.8	2.6	2.6	0.5	0.6	0.7	0.7	
Net FDI (negative = inflow)	**-1.3**	**-1.2**	**-1.1**	**-2.0**	**1.6**	**-1.8**	**-2.2**	**-2.6**	**-2.9**	**-3.2**	**-3.5**		**-3.5**	**-3.3**	**-3.3**	**-3.3**	**-3.2**	**-2.5**	**-2.5**	-3.0
Endogenous debt dynamics 2/	**0.1**	**-1.1**	**-0.5**			**-1.0**	**-1.4**	**-1.4**	**-1.5**	**-1.5**	**-1.4**		**-1.2**	**-0.9**	**-0.8**	**-0.8**	**-0.7**	**-0.6**	**-0.6**	
Contribution from nominal interest rate	3.0	0.2	0.2			0.2	0.3	0.3	0.2	0.3	0.3		0.4	0.5	0.6	0.6	0.6	0.1	0.1	
Contribution from real GDP growth	-1.5	-1.2	-0.8			-1.2	-1.7	-1.6	-1.7	-1.7	-1.6		-1.6	-1.5	-1.4	-1.4	-1.3	-0.7	-0.7	
Contribution from price and exchange rate changes	-1.4	-0.1	0.1			…	…	…	…	…	…		…	…	…	…	…	…	…	
Residual (3-4) 3/	**-2.0**	**0.1**	**0.3**			**5.3**	**0.9**	**2.2**	**2.5**	**2.3**	**1.8**		**1.2**	**0.2**	**0.6**	**2.4**	**1.8**	**0.9**	**1.8**	
o/w exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	0.0	
PV of external debt 4/	...	...	15.9			22.4	21.4	21.1	20.6	19.9	19.1		18.3	16.9	16.1	15.3	14.4	9.2	8.7	
In percent of exports	...	...	61.0			76.6	64.4	63.4	61.4	59.0	56.9		54.4	52.4	49.6	46.6	43.6	25.5	23.7	
PV of PPG external debt	**...**	**...**	**15.9**			**22.4**	**21.4**	**21.1**	**20.6**	**19.9**	**19.1**		**18.3**	**16.9**	**16.1**	**15.3**	**14.4**	**9.2**	**8.7**	
In percent of exports	**...**	**...**	**61.0**			**76.6**	**64.4**	**63.4**	**61.4**	**59.0**	**56.9**		**54.4**	**52.4**	**49.6**	**46.6**	**43.6**	**25.5**	**23.7**	
In percent of government revenues	**...**	**...**	**60.5**			**104.8**	**96.8**	**89.9**	**86.8**	**82.7**	**76.9**		**72.5**	**67.0**	**61.3**	**56.3**	**51.6**	**25.7**	**23.8**	
Debt service-to-exports ratio (in percent)	**15.2**	**4.3**	**1.9**			**2.4**	**2.5**	**3.6**	**3.6**	**3.6**	**3.2**		**3.7**	**3.5**	**3.8**	**3.8**	**3.6**	**1.7**	**1.6**	
PPG debt service-to-exports ratio (in percent)	**15.2**	**4.3**	**1.9**			**2.4**	**2.5**	**3.6**	**3.6**	**3.6**	**3.2**		**3.7**	**3.5**	**3.8**	**3.8**	**3.6**	**1.7**	**1.6**	
PPG debt service-to-revenue ratio (in percent)	**14.9**	**4.1**	**1.8**			**3.3**	**3.8**	**5.1**	**5.1**	**5.1**	**4.3**		**4.9**	**4.5**	**4.6**	**4.6**	**4.3**	**1.7**	**1.6**	
Total gross financing need (Billions of U.S. dollars)	0.2	0.1	0.3			0.1	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.1	-0.1	-0.1	-0.2	-0.3	
Non-interest current account deficit that stabilizes debt ratio	3.2	2.2	1.3			-2.6	2.6	1.7	1.9	2.4	3.1		3.4	4.1	3.5	1.7	2.1	2.2	1.2	
Key macroeconomic assumptions																				
Real GDP growth (in percent)	9.0	6.5	4.3	5.8	2.7	4.3	5.7	6.1	6.5	6.7	6.7	6.0	6.5	6.5	6.6	6.6	6.1	5.2	5.3	5.8
GDP deflator in US dollar terms (change in percent)	7.9	0.6	-0.4	2.9	7.2	-19.1	-3.3	1.3	0.2	1.0	1.5	-3.0	1.3	5.2	0.9	0.3	1.1	6.1	2.6	2.4
Effective interest rate (percent) 5/	18.1	0.9	1.1	2.8	5.4	0.6	1.1	1.0	0.9	1.0	1.0	0.9	1.7	2.3	2.8	2.8	2.7	1.0	1.0	1.7
Growth of exports of G&S (US dollar terms, in percent)	0.7	29.5	3.6	14.0	15.9	-1.6	15.7	7.9	8.0	8.0	8.0	7.7	8.0	7.7	7.9	8.1	8.2	10.1	10.3	8.9
Growth of imports of G&S (US dollar terms, in percent)	-4.7	19.7	-7.2	10.3	16.0	1.9	4.1	5.9	5.7	6.8	7.8	5.4	7.2	7.4	7.5	7.6	7.6	8.1	8.2	7.8
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	39.5	39.5	39.5	39.5	42.0	42.0	40.4	42.0	42.0	42.0	42.0	42.0	42.0	42.0	42.0
Government revenues (excluding grants, in percent of GDP)	21.3	26.5	26.3			21.4	22.1	23.4	23.8	24.0	24.9		25.2	25.3	26.2	27.1	28.0	35.7	36.6	30.7
Aid flows (in Billions of US dollars) 7/	0.4	0.6	0.2			0.5	0.5	0.5	0.5	0.5	0.5		0.5	0.5	0.5	0.5	0.6	0.7	0.6	
o/w Grants	0.4	0.6	0.2			0.4	0.4	0.4	0.4	0.4	0.4		0.4	0.4	0.4	0.4	0.5	0.5	0.5	
o/w Concessional loans	0.0	0.0	0.0			0.1	0.1	0.1	0.1	0.1	0.1		0.1	0.1	0.1	0.1	0.1	0.1	0.1	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			10.0	10.2	9.3	8.6	8.1	7.4		6.9	6.3	5.9	5.6	5.3	3.1	2.7	4.6
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			87.8	84.9	86.1	85.9	86.5	86.5		86.5	86.5	86.4	86.4	86.4	85.7	84.9	86.1
Memorandum items:																				
Nominal GDP (Billions of US dollars)	4.3	5.0	5.4			4.7	4.8	5.2	5.6	6.0	6.5		7.0	7.8	8.4	9.0	9.7	19.8	21.4	
Nominal dollar GDP growth	17.3	17.7	7.2			-12.3	2.2	7.5	6.8	7.8	8.3	3.4	7.9	12.0	7.5	6.9	7.3	11.7	8.0	8.3
PV of PPG external debt (in Billions of US dollars)			0.9			0.9	1.0	1.1	1.1	1.2	1.2		1.3	1.3	1.3	1.3	1.4	1.8	1.8	
(PVt-PVt-1)/GDPt-1 (in percent)						1.2	1.8	1.2	1.0	0.9	0.9	1.2	0.6	0.6	0.4	0.3	0.2	0.3	0.2	0.4
Gross workers' remittances (Billions of US dollars)	…	…	…			…	…	…	…	…	…		…	…	…	…	…	…	…	
PV of PPG external debt (in percent of GDP + remittances)	...	...	15.9			22.4	21.4	21.1	20.6	19.9	19.1		18.3	16.9	16.1	15.3	14.4	9.2	8.7	
PV of PPG external debt (in percent of exports + remittances)	...	...	61.0			76.6	64.4	63.4	61.4	59.0	56.9		54.4	52.4	49.6	46.6	43.6	25.5	23.7	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	1.9			2.4	2.5	3.6	3.6	3.6	3.2		3.7	3.5	3.8	3.8	3.6	1.7	1.6	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2.Malawi: Public Sector Debt Sustainability Framework, Baseline Scenario, 2009-2032
(In percent of GDP, unless otherwise indicated)

	Actual													Estimate					Projections				
	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	Average 5/	Standard Deviation 5/	2012	2013	2014	2015	2016	2017	2012-17 Average	2022	2032	2018-32 Average
Public sector debt 1/	...	...	179.1	165.3	146.7	39.2	41.0	48.5	49.5	39.4	48.3			52.7	48.4	45.4	43.1	40.6	38.8		29.2	18.9	
o/w foreign-currency denominated	...	...	160.3	145.5	127.9	22.6	19.0	19.5	19.4	18.4	23.0			29.7	28.4	28.2	27.7	26.8	25.9		21.8	13.7	
Change in public sector debt	...	...	...	-13.8	-18.6	-107.5	1.8	7.5	1.0	-10.1	8.9			4.4	-4.3	-3.1	-2.3	-2.5	-1.8		-1.2	-0.9	
Identified debt-creating flows		...	...	8.2	-15.3	-21.5	1.3	0.0	3.2	-7.0	3.3			11.5	-5.4	-3.6	-2.7	-3.0	-2.3		-1.3	-1.6	
Primary deficit	...	...	-1.6	-3.1	-3.0	-5.9	2.1	3.2	-0.6	-4.7	2.3	-1.2	3.3	3.2	-0.1	-1.5	-0.7	-0.8	-0.1	0.0	-0.4	-0.7	-0.3
Revenue and grants	...	...	27.6	32.8	33.3	41.9	37.4	36.5	35.2	42.0	31.9			30.5	31.0	31.8	31.5	31.2	31.5		32.7	38.9	
of which: grants	...	...	9.1	12.4	11.4	18.9	15.3	13.1	10.2	13.6	4.5			9.1	9.0	8.3	7.7	7.1	6.6		4.7	2.3	
Primary (noninterest) expenditure	...	...	26.0	29.7	30.3	36.0	39.6	39.7	34.6	37.3	34.2			33.7	31.0	30.3	30.8	30.4	31.4		32.3	38.2	
Automatic debt dynamics		...	...	11.3	-12.3	-14.7	-0.8	-3.2	-0.9	-2.3	0.9			8.3	-5.3	-2.1	-2.0	-2.2	-2.2		-0.9	-0.9	
Contribution from interest rate/growth differential	...	...	...	-5.2	-3.2	-5.1	-1.9	-2.2	0.5	-1.9	0.6			-2.4	-3.3	-2.1	-2.2	-2.2	-2.2		-1.0	-0.8	
of which: contribution from average real interest rate	...	...	...	3.0	1.3	-2.1	1.5	0.9	4.5	1.1	2.3			-0.4	-0.4	0.7	0.6	0.5	0.3		0.7	0.2	
of which: contribution from real GDP growth	...	...	...	-8.3	-4.6	-3.0	-3.4	-3.2	-4.0	-3.0	-1.6			-2.0	-2.9	-2.8	-2.8	-2.7	-2.5		-1.8	-1.0	
Contribution from real exchange rate depreciation		...	...	16.5	-9.1	-9.6	1.1	-0.9	-1.4	-0.4	0.3			10.7	-2.0	0.0	0.1	0.0	0.0		...	...	
Other identified debt-creating flows		0.0	0.0	0.0	0.0	-1.0	-0.1	0.0	4.8	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)		...	0.0	0.0	0.0	-0.4	0.0	0.0	-0.1	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities		...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)		...	0.0	0.0	0.0	-0.6	-0.1	0.0	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)		...	0.0	0.0	0.0	0.0	0.0	0.0	4.8	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes		...	...	-21.9	-3.3	-86.0	0.5	7.5	-2.2	-3.1	5.6			-7.1	1.1	0.5	0.4	0.5	0.5		0.1	0.6	
Other Sustainability Indicators																							
PV of public sector debt			...	...	...	...	...	...	...	...	41.2			45.5	41.4	38.3	36.1	33.7	32.0		21.9	13.9	
o/w foreign-currency denominated			...	...	...	...	...	...	...	...	15.9			22.4	21.4	21.1	20.6	19.9	19.1		14.4	8.7	
o/w external		...	...	...	...	...	...	...	...	...	15.9			22.4	21.4	21.1	20.6	19.9	19.1		14.4	8.7	
PV of contingent liabilities (not included in public sector debt)		...	...	...	...	...	...	...	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	...	...	21.0	25.3	23.2	18.8	18.4	17.8	25.2	20.0	20.0			21.7	18.9	16.7	15.2	13.5	12.7		7.9	5.1	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	...	...	...	...	...	...	...	...	129.3			149.2	133.4	120.5	114.7	107.9	101.8		67.0	35.8	
PV of public sector debt-to-revenue ratio (in percent)	...	...	...	...	...	...	...	...	...	...	150.8			212.5	187.6	163.4	151.9	140.0	128.8		78.3	38.1	
o/w external 3/		...	...	...	...	...	...	...	...	...	58.3			104.8	96.8	89.9	86.8	82.7	76.9		51.6	23.8	
Debt service-to-revenue and grants ratio (in percent) 4/		35.3	33.8	35.8	28.6	18.8	9.3	8.4	19.5	9.5	10.5			8.5	8.4	9.5	8.7	8.1	7.0		6.4	3.1	
Debt service-to-revenue ratio (in percent) 4/		46.9	50.4	57.6	43.5	34.2	15.6	13.1	27.4	14.0	12.2			12.2	11.8	12.8	11.5	10.5	8.8		7.5	3.3	
Primary deficit that stabilizes the debt-to-GDP ratio									-1.6	5.4	-6.6			-1.2	4.2	1.6	1.6	1.7	1.7		0.8	0.2	
Key macroeconomic and fiscal assumptions																							
Real GDP growth (in percent)	...	...	5.1	4.9	2.8	2.1	9.5	8.3	9.0	6.5	4.3	5.8	2.7	4.3	5.7	6.1	6.5	6.7	6.7	6.0	6.1	5.3	5.8
Average nominal interest rate on forex debt (in percent)		0.8	0.8	0.9	0.7	0.6	1.2	0.8	20.4	1.0	1.4	2.9	6.2	0.7	1.2	1.1	0.9	1.1	1.1	1.0	2.8	1.1	1.7
Average real interest rate on domestic debt (in percent)		...	48.0	32.9	25.4	6.2	12.2	5.7	4.1	4.1	12.0	16.7	15.4	-1.6	-2.6	3.4	4.4	4.1	3.9	1.9	7.2	6.2	6.8
Real exchange rate depreciation (in percent, + indicates depreciation)		...	20.2	11.0	-6.6	-7.9	5.3	-5.3	-6.6	-2.2	1.6	1.1	9.6	48.8	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	...	...	10.0	15.4	11.0	27.3	10.0	8.7	8.4	7.4	3.8	11.3	6.8	16.4	14.2	7.3	6.0	5.8	5.6	9.2	6.2	6.2	6.7
Growth of real primary spending (deflated by GDP deflator, in percent)		...	0.1	0.1	0.1	0.1	0.3	0.1	-0.1	0.2	0.0	0.1	0.1	0.1	0.0	0.0	0.1	0.1	0.1	0.1	0.1	0.0	0.1
Grant element of new external borrowing (in percent)		...	...	...	...	...	...	...	...	...	...	...	...	39.5	39.5	39.5	39.5	42.0	42.0	40.4	42.0	42.0	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 3.Malawi: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012-2032
(In percent)

	Projections											
	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	**2022**	2032
PV of debt-to GDP ratio												
Baseline	22	21	21	21	20	19	18	17	16	16	**15**	8
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2012-2032 1/	22	23	26	29	32	34	36	38	39	41	**43**	50
A2. New public sector loans on less favorable terms in 2012-2032 2	22	22	22	22	22	22	22	20	20	20	**19**	14
B. Bound Tests												
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	22	21	22	21	21	20	19	18	17	16	**15**	9
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	22	26	35	34	33	32	30	28	27	26	**24**	12
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	22	21	22	22	21	20	19	18	17	16	**15**	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	22	25	28	28	27	26	25	23	22	21	**20**	10
B5. Combination of B1-B4 using one-half standard deviation shocks	22	24	29	28	28	27	25	24	22	22	**20**	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	22	30	30	29	28	27	26	24	23	22	**21**	12
PV of debt-to-exports ratio												
Baseline	77	64	63	61	59	57	55	53	50	47	**44**	23
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2012-2032 1/	77	69	78	86	94	102	107	117	120	125	**129**	135
A2. New public sector loans on less favorable terms in 2012-2032 2	77	65	66	66	66	65	64	63	62	60	**57**	37
B. Bound Tests												
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	77	63	62	60	58	56	54	52	49	47	**44**	23
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	77	110	185	179	172	166	159	153	146	138	**130**	56
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	77	63	62	60	58	56	54	52	49	47	**44**	23
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	77	75	85	82	79	77	74	71	67	64	**60**	27
B5. Combination of B1-B4 using one-half standard deviation shocks	77	81	95	92	89	86	82	79	75	71	**67**	31
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	77	63	62	60	58	56	54	52	49	47	**44**	23
PV of debt-to-revenue ratio												
Baseline	105	97	90	87	83	77	73	68	62	57	**53**	23
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2012-2032 1/	105	104	111	122	132	138	143	149	148	151	**153**	136
A2. New public sector loans on less favorable terms in 2012-2032 2	105	98	94	94	92	88	86	81	76	72	**68**	37
B. Bound Tests												
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	105	97	92	89	86	80	76	70	64	59	**54**	24
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	105	120	149	143	137	127	120	111	102	95	**87**	32
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	105	95	94	91	87	81	77	71	65	60	**55**	24
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	105	113	120	116	111	104	98	90	83	77	**70**	27
B5. Combination of B1-B4 using one-half standard deviation shocks	105	110	124	120	115	107	101	93	86	79	**73**	28
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	105	137	127	123	118	110	104	96	88	81	**75**	33

Table 3.Malawi: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012-2032 (continued)

(In percent)

Debt service-to-exports ratio

Baseline	2	3	4	4	4	3	4	3	4	4	**4**	2
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2012-2032 1/	2	2	4	4	4	3	4	4	4	4	**5**	6
A2. New public sector loans on less favorable terms in 2012-2032 2	2	3	4	4	4	4	4	4	4	4	**4**	2
B. Bound Tests												
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	2	3	4	4	4	3	4	3	4	4	**4**	2
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	2	3	7	8	7	7	7	7	7	7	**8**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	2	3	4	4	4	3	4	3	4	4	**4**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	2	3	4	4	4	3	4	4	4	4	**4**	2
B5. Combination of B1-B4 using one-half standard deviation shocks	2	3	4	4	4	4	4	4	4	4	**5**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	2	3	4	4	4	3	4	3	4	4	**4**	2

Debt service-to-revenue ratio

Baseline	3	4	5	5	5	4	5	4	5	5	**4**	2
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2012-2032 1/	3	4	5	5	5	5	5	5	5	5	**6**	6
A2. New public sector loans on less favorable terms in 2012-2032 2	3	4	5	5	5	5	5	5	5	5	**5**	2
B. Bound Tests												
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	3	4	5	5	5	5	5	5	5	5	**5**	2
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	3	4	6	6	6	5	6	5	5	5	**6**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	3	4	5	5	5	5	5	5	5	5	**5**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	3	4	5	6	6	5	5	5	5	5	**5**	2
B5. Combination of B1-B4 using one-half standard deviation shocks	3	4	5	6	6	5	5	5	5	5	**5**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	3	5	7	7	7	6	7	6	7	7	**6**	3
Memorandum item:												
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	38	38	38	38	38	38	38	38	38	38	**38**	38

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 4.Malawi: Sensitivity Analysis for Key Indicators of Public Debt 2012-2032

	Projections							
	2012	2013	2014	2015	2016	2017	2022	2032
PV of Debt-to-GDP Ratio								
Baseline	45	41	38	36	34	32	22	14
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	45	40	37	35	33	31	20	7
A2. Primary balance is unchanged from 2012	45	43	43	43	42	42	41	41
A3. Permanently lower GDP growth 1/	45	41	38	37	35	34	27	34
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	45	42	41	39	38	37	30	27
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	45	42	41	39	36	35	24	15
B3. Combination of B1-B2 using one half standard deviation shocks	45	42	40	38	36	35	26	21
B4. One-time 30 percent real depreciation in 2013	45	49	46	43	40	38	26	17
B5. 10 percent of GDP increase in other debt-creating flows in 2013	45	47	44	41	39	37	26	16
PV of Debt-to-Revenue Ratio 2/								
Baseline	138	133	121	115	108	102	67	36
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	149	129	118	112	105	98	61	18
A2. Primary balance is unchanged from 2012	149	138	134	135	136	134	125	106
A3. Permanently lower GDP growth 1/	149	132	121	116	111	107	83	86
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	149	135	127	124	119	115	90	70
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	149	136	129	123	117	110	74	39
B3. Combination of B1-B2 using one half standard deviation shocks	149	134	126	121	116	111	80	54
B4. One-time 30 percent real depreciation in 2013	149	159	144	136	128	120	80	43
B5. 10 percent of GDP increase in other debt-creating flows in 2013	149	151	137	131	124	117	79	42
Debt Service-to-Revenue Ratio 2/								
Baseline	9	8	9	9	8	7	6	3
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	11	10	10	9	8	7	7	3
A2. Primary balance is unchanged from 2012	11	10	10	9	9	8	8	6
A3. Permanently lower GDP growth 1/	11	10	10	9	9	7	7	5
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	11	10	10	10	9	8	7	4
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	11	10	10	9	9	7	7	3
B3. Combination of B1-B2 using one half standard deviation shocks	11	10	10	9	9	8	7	4
B4. One-time 30 percent real depreciation in 2013	11	10	12	11	10	9	9	4
B5. 10 percent of GDP increase in other debt-creating flows in 2013	11	10	10	9	9	8	7	4

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.